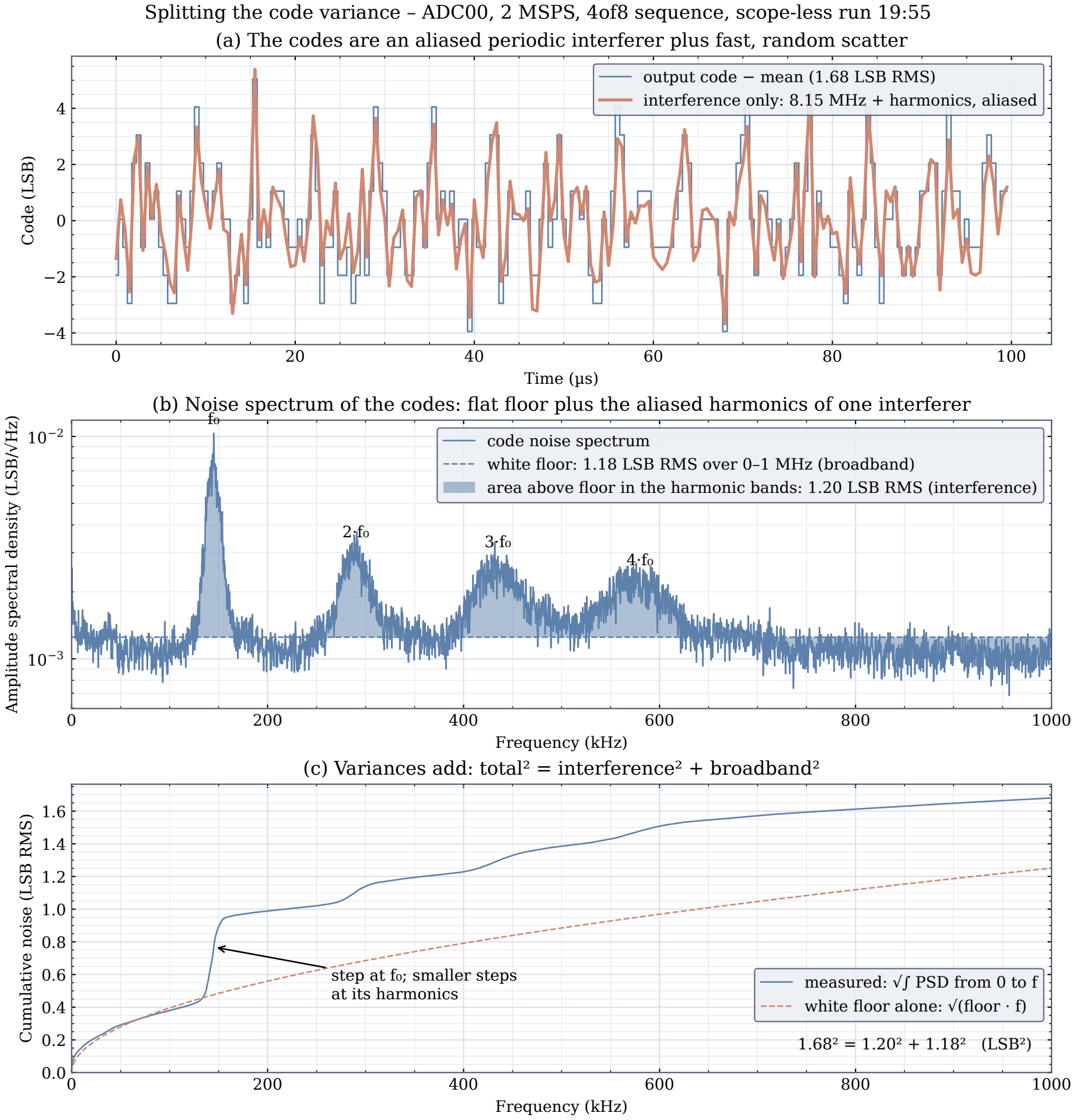

Splitting the code variance – ADC00, 2 MSPS, 4of8 sequence, scope-less run 19:55
(a) The codes are an aliased periodic interferer plus fast, random scatter
output code − mean (1.68 LSB RMS)
interference only: 8.15 MHz + harmonics, aliased
Code (LSB)
Time (µs)
(b) Noise spectrum of the codes: flat floor plus the aliased harmonics of one interferer
code noise spectrum
white floor: 1.18 LSB RMS over 0–1 MHz (broadband)
area above floor in the harmonic bands: 1.20 LSB RMS (interference)
f₀
2·f₀
3·f₀
4·f₀
Amplitude spectral density (LSB/√Hz)
Frequency (kHz)
(c) Variances add: total² = interference² + broadband²
step at f₀; smaller steps
at its harmonics
measured: √∫ PSD from 0 to f
white floor alone: √(floor · f)
1.68² = 1.20² + 1.18² (LSB²)
Cumulative noise (LSB RMS)
Frequency (kHz)